

◆ 16.4 Conservative Mode (Quality-Adaptive Execution — Final Version)

Core Principle

Under uncertainty or degraded quality, execution is restricted, not merely disabled.

16.4.1 Global Quality Index

$$\begin{aligned} Q(t) = & \\ & w_I I_{\{\mathrm{eff}\}} \\ & + w_T (1 - C_{\{\mathrm{time}\}}) \\ & + w_S (1 - C_{\{\mathrm{struct}\}}) \\ & + w_L (1 - C_{\{\mathrm{loop}\}}) \end{aligned}$$

$$Q(t) \in [0,1], \quad \sum w_i = 1$$

16.4.2 Operating Regimes

● Normal Mode

$$Q \geq \theta_{\{\mathrm{high}\}} \\ \rightarrow$$

$\gamma = 1, \quad u \in \mathcal{U}_{\mathrm{safe}}$

🟡 Conservative Mode

$\theta_{\mathrm{low}} \leq Q < \theta_{\mathrm{high}}$
 $\rightarrow$
 $\gamma = 1, \quad u \in \alpha(Q), \mathcal{U}_{\mathrm{safe}}$

$0 < \alpha(Q) < 1$

🔴 Failure Mode

$Q < \theta_{\mathrm{low}}$
 $\rightarrow$
 $\gamma = 0$

16.4.3 Control Scaling

$u_{\mathrm{eff}} = \alpha(Q)u$

$$\alpha(Q) = \frac{Q - \theta_{\mathrm{low}}}{\theta_{\mathrm{high}} - \theta_{\mathrm{low}}}$$

$$\alpha(Q) \in [0,1]$$

16.4.4 Stability Preservation

$$\begin{aligned} & \nabla V \cdot u_{\mathrm{eff}} \\ &= \\ & \alpha(Q) \nabla V \cdot u \\ & \leq \\ & \alpha(Q) B \\ & \leq B \end{aligned}$$

Thus:

stability constraints remain satisfied under scaling.

16.4.5 Operational Rules

- Decreasing quality $\Rightarrow$ gradual reduction before shutdown

- No escalation in conservative mode
- Recovery is smooth as Q improves
- Execution always remains within

$\mathcal{U}_{\mathrm{safe}}$

Key Insight

The system degrades continuously, not abruptly, while preserving stability.

◆ 16.5 Noise and Stress Impact (Adaptive Robustness)

Core Principle

Increased noise reduces confidence and tightens decision thresholds.

16.5.1 Noise Intensity

$\sigma_n(t)$

$\quad \text{(instantaneous noise estimate)}$

16.5.2 Normalized Noise Level

$$N(t) = \min_{\lambda \in [0,1]} \left(1, \frac{\sigma_n(t)}{\sigma_{\mathrm{ref}}} \right)$$

16.5.3 Impact on Independence

$$I_{\mathrm{eff}}^* = I_{\mathrm{eff}} \big(1 - \lambda_n N \big)$$
$$\lambda_n \in [0,1]$$

16.5.4 Threshold Tightening

$$\theta_{\mathrm{ind}}^* = \theta_{\mathrm{ind}} + \kappa_n N$$
$$\kappa_n \geq 0$$

16.5.5 Impact on Execution

u_{eff}

=

$\alpha(Q), (1 - N), u$

16.5.6 Decision Effect

$N \uparrow$

$\rightarrow$

$I_{\mathrm{eff}}^* \downarrow$,

$\quad$

$\theta_{\mathrm{ind}}^* \uparrow$

$\rightarrow$

$\gamma \rightarrow 0$

16.5.7 Operational Rules

- High noise $\Rightarrow$ reduce or suppress execution
- Continuous re-estimation (rolling noise estimation)
- Low SNR channels must be down-

weighted or excluded

Key Insight

Noise simultaneously reduces confidence and raises decision thresholds.

Final Insight (Very Important

- **Epistemic reliability (Q , I_{eff})**
- **Physical execution (u scaling)**
- **Robustness (noise adaptation)**

self-aware + self-regulating

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
DOI: 10.5281/zenodo.19521219